I Multiple choice questions

1. The dimension of $\frac{1}{\mu_o \epsilon_o}$ is

(a) [L T^{-1}] (b) [L^2 T^{-2}]

(c) [L^{-1} T] (d) [L^{-2} T^2]

2. If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a electromagnetic waves is

(a) 100 V m^{-1} (b) 300 V m^{-1}

(c) 600 V m^{-1} (d) 900 V m^{-1}

3. Which of the following electromagnetic radiations is used for viewing objects through fog

(a) microwave (b) gamma rays

(c) X- rays (d) infrared

4. Which of the following is false for electromagnetic waves

(a) transverse

(b) non-mechanical waves

(c) longitudinal

(d) produced by accelerating charges

5. Consider an oscillator which has a charged particle oscillating about its mean position with a frequency of 300 MHz. The wavelength of electromagnetic waves produced by this oscillator is

(a) 1 m (b) 10 m

(c) 100 m (d) 1000 m

6. The electric and the magnetic fields, associated with an electromagnetic wave, propagating along negative X axis can be represented by

(a) $\vec{E} = E_o \hat{i}$ and $\vec{B} = B_o \hat{k}$

(b) $\vec{E} = E_o \hat{k}$ and $\vec{B} = B_o \hat{j}$

(c) $\vec{E} = E_o \hat{i}$ and $\vec{B} = B_o \hat{j}$

(d) $\vec{E} = E_o \hat{j}$ and $\vec{B} = B_o \hat{i}$

7. In an electromagnetic wave travelling in free space the rms value of the electric field is 3 V m^{-1}. The peak value of the magnetic field is

(a) 1.414×10^{-8} T (b) 1.0×10^{-8} T

(c) 2.828×10^{-8} T (d) 2.0×10^{-8} T

8. An e.m. wave is propagating in a medium with a velocity $\vec{v} = v\hat{i}$. The instantaneous oscillating electric field of this e.m. wave is along +y-axis, then the direction of oscillating magnetic field of the e.m. wave will be along:

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(a) –y direction

(b) –x direction

(c) +z direction

(d) –z direction

9. If the magnetic monopole exists, then which of the Maxwell's equation to be modified?.

(a) $\oint_s \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_o}$

(b) $\oint_s \vec{B} \cdot d\vec{A} = 0$

(c) $\oint_l \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d}{dt} \oint_s \vec{E}.\vec{dA}$

(d) $\oint_l \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\Phi_B$

10. Fraunhofer lines are an example of _____ spectrum.

(a) line emission (b) line absorption

(c) band emission (d) band absorption

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